

Presentation Preparation

Sim-to-Real with TransformerPayne – AS4501

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1 Presentation Preparation — Sim-to-Real with TransformerPayne

1.0.1 Everything you need to defend this work

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1.1 PART I — THE STORY IN ONE PAGE

1.1.1 The question

Radiative-transfer codes can compute a stellar spectrum from physics, but they are far too slow to generate the millions of spectra a machine-learning model wants. **Spectral emulators** (The Payne, TransformerPayne) are neural networks trained to imitate those codes: give them stellar labels (temperature, gravity, chemical abundances) and they emit a spectrum in milliseconds.

That raises the question this project exists to answer:

Can we train a classifier entirely on emulator-generated (synthetic) spectra, and have it work on real telescope data?

This is a **sim-to-real** problem. If it works, we get unlimited, perfectly-labelled training data for free. If it fails, we have learned something about where the simulation departs from reality.

1.1.2 The answer

Yes — and on a well-behaved instrument it works perfectly.

	DESI	SDSS
Trained on synthetic → tested on real	0.752	0.956
Trained on real → tested on real (ceiling)	0.843	0.940

	DESI	SDSS
Domain shift = ceiling – synthetic	+0.090	–0.017
Cross-survey: real DESI → real SDSS	—	0.828

On **SDSS**, the model trained on *nothing but synthetic spectra* (0.956) matches — even slightly exceeds — a model trained on **real, labelled** SDSS spectra (0.940). The domain shift is **zero**.

And note the asymmetry in effort: the synthetic model had **10,000** training spectra; the real model had only **594**, because that is all the labelled data there is. **The synthetic approach gives you more data, needs no labels, and loses nothing**. That is the thesis of the project, demonstrated.

1.1.3 Why DESI looks worse — and why that is not our failure

DESI gives 0.752. But **DESI's own ceiling is 0.843** — no model, however good, does better on DESI's data with DESI's labels (SDSS's ceiling is 0.940). And a model trained on **real DESI** reaches only **0.828** on **real SDSS**: the two instruments disagree *with each other*, with no simulation involved at all.

The measured cause: comparing the mean spectrum of G stars *at the same temperature* in both surveys, every line agrees within 3–10 % — **except H-beta, which is 20 % shallower in DESI**. And SHAP shows the classifier decides almost entirely with H-beta. Weak H-beta → weak Balmer → looks cool → **K**. That is exactly the observed error: DESI's G stars get called K, at every temperature, and almost never the reverse.

1.2 PART II — THE THEORY YOU NEED

1.2.1 1. Why spectral type = temperature

A star's spectral class (O B A F G K M) is, at bottom, a **temperature sequence** — O hottest (>30,000 K), M coolest (~3,000 K). Our two classes:

- **G** : 5,300–6,000 K (the Sun is G2, 5,772 K)
- **K** : 4,000–5,200 K

The boundary sits at ~**5,250 K**. So G vs K is a *temperature cut* — remember this, it explains everything downstream.

1.2.2 2. Balmer lines are the thermometer

Hydrogen is the most abundant element, and its **Balmer series** (H-beta 4862 Å, H-gamma 4341 Å, H-delta 4103 Å) arises from the n=2 level. Populating n=2 requires thermal excitation, so:

- **Hotter star (G)** → more atoms in n=2 → **stronger, deeper Balmer absorption**
- **Cooler star (K)** → fewer atoms in n=2 → **weak Balmer**

Balmer strength therefore tracks temperature *directly*, and — crucially — it is **almost independent of chemical composition**, because hydrogen abundance barely varies between stars.

Meanwhile, **metal lines** (Ca I 4227 Å, the CH G-band at 4300 Å, hundreds of Fe lines) get stronger in cooler stars *but also* depend on how metal-rich the star is. They are **ambiguous**: a metal-rich G star can look, in Ca I, like a metal-poor K star.

Expect the question: “Why does your model use H-beta and not the metal lines?”

Answer: Because we randomize abundances ([X/H] from –1 to +1 dex) in the training set, the metal lines carry a confounded signal, while Balmer is a clean thermometer. The Random Forest discovered this by itself — see Part III.

1.2.3 3. The two emulators

	The Payne (Ting et al. 2019)	TransformerPayne (Róžański et al.)
Architecture	fully-connected MLP	transformer with attention
Input	stellar labels	stellar labels + the wavelength itself
Output	the whole spectrum at once	the flux at that wavelength
Key idea	fast surrogate of a physics code	wavelength as a <i>parameter</i> → resolution-independent, and attention can associate lines with elements

Both are **emulators**, not physics: they are trained on a library of spectra computed by a real radiative-transfer code. They are fast (ms vs hours) but only valid **inside the parameter grid they were trained on** — for TransformerPayne, roughly **dwarfs at 4,000–6,000 K**. This validity limit is the single most important caveat in the whole project (see Part V).

1.2.4 4. Preprocessing — and why it is not a detail

Real spectra and synthetic spectra do not live in the same space. Three operations must be applied **identically to both** or the classifier learns the difference between the two datasets instead of the difference between G and K stars.

(a) Common wavelength grid. Everything is resampled onto 4000–5000 Å, 1000 pixels. This range is where TransformerPayne is validated *and* where DESI’s data lives.

(b) Instrumental broadening (LSF). A real spectrograph smears every line. Its **resolving power** is $R = \lambda/\Delta\lambda$. Synthetic spectra come at $R = 100,000$ (razor-sharp); DESI is $R \approx 2,000$ – $2,700$, SDSS $R \approx 2,000$. So we **convolve the synthetic spectra with a Gaussian** to bring them down to the instrument’s resolution. DESI’s R varies with wavelength, so we do this with a coordinate warp that makes a constant- σ convolution produce the correct wavelength-dependent FWHM. Equivalent width is conserved — the lines get broader and shallower, not weaker.

(c) Continuum normalization. The absolute flux depends on the star’s distance and brightness, which tell us nothing about its type. So we divide out the smooth **continuum**, leaving only the line pattern. We fit the continuum with a low-order polynomial using **asymmetric sigma clipping** (reject points far *below* the fit — those are absorption lines — keep refitting), so the fit tracks the upper envelope.

Expect the question: “Why does the continuum normalization matter so much?”

Answer: Because it determines the *apparent depth* of every line. Broad lines like H-beta have wings spanning ± 40 Å; a continuum fit can partially absorb those wings, and how much it absorbs depends on the noise. That is precisely how DESI’s H-beta ends up 20 % shallower than SDSS’s.

1.2.5 5. Sim-to-real and the domain shift

The vocabulary from domain adaptation, which you should use:

- **Source-only** — train on synthetic, test on real. *This is our method.*
- **Target-supervised** — train on real labelled data, test on real. *This is the ceiling.*
- **Domain shift** — the gap between them. **By definition.**

This is the most important methodological point in your talk. A single number like “0.76 on DESI” is meaningless without the ceiling. Good or bad *compared to what?* Reporting source-only together with target-supervised is standard practice in every domain-adaptation paper, and the gap between them *is* the domain shift. It is what turns your result from a number into a claim.

1.2.6 6. Random Forest, Gini, and SHAP

A **Random Forest** is an ensemble of decision trees; each tree splits on pixel values. It is a good choice here because it is robust, needs no scaling, and — the point — is **interpretable**.

- **Gini importance** — how much each pixel reduces impurity across the trees. Built-in, but biased toward correlated / high-cardinality features.
- **SHAP** (SHapley Additive exPlanations) — from cooperative game theory: it distributes the prediction fairly among the features, with axioms (efficiency, symmetry) that guarantee consistency. For tree models it is computed **exactly** (TreeExplainer), not approximated.

We report both; they agree, which is itself evidence the result is not an artifact of one method.

1.2.7 7. The Initial Mass Function (why hot stars are rare)

Star formation produces far more low-mass stars than high-mass ones (the IMF, $dN/dM \sim M^{-2.35}$ for massive stars). Massive O and B stars are also short-lived. Consequence: **they barely exist**. LAMOST — one of the largest spectroscopic surveys ever — contains **234 O stars** in total, against **3.5 million G stars**. Any “balanced sample over all spectral classes” is therefore capped by the rarest class. This is physics, not a data-collection failure.

1.3 PART III — THE RESULTS, IN THE ORDER YOU SHOULD PRESENT THEM

1.3.1 Result 1 — The domain shift, and how we closed it

Stage	Real DESI accuracy
Baseline (sharp synthetic, percentile normalization)	0.66
+ $R(\lambda)$ instrumental broadening	0.67
+ improved continuum normalization <i>alone</i>	0.57
+ both together	0.74

The lesson: the two effects are coupled. Broadening alone barely helps. The better normalization *alone* makes things **worse** — because the polynomial continuum reacts differently to sharp synthetic lines than to broad real ones, so it normalizes the two datasets *differently*. Only when the synthetic spectra are first broadened to the real resolution does the same continuum fit treat both consistently. **Match the resolution first, then the normalization.**

This also fixed the original failure mode: K stars used to be misclassified as G at a rate of 57 % (recall 0.43); afterwards K recall is 0.85.

1.3.2 Result 2 — Scaling saturates (this answers the professor’s 100k)

Synthetic training spectra	Real accuracy
2,000	0.776 ± 0.013
10,000	0.752 ± 0.004
50,000	0.761 ± 0.006
100,000	0.759 ± 0.007

Flat from 2,000 onward. More synthetic data buys nothing. The limit is not sample size. This is also the answer to the team’s crisis about being unable to obtain 700,000 spectra per class: *it does not matter*. We can prove it with a curve instead of apologizing for it.

1.3.3 Result 3 — The emulator is not the bottleneck

We trained a **Payne-style MLP** on TransformerPayne spectra (reconstruction RMSE 0.011 — below the injected noise level of 0.02, so effectively exact) and compared the sim→real transfer:

- TransformerPayne: **0.78**
- The Payne (MLP): **0.77**

Statistically identical. **Architecture is not the problem.**

Be honest about this: our Payne is trained on TransformerPayne output, so it is a fast **surrogate**, not an independent physical check. Present it as “second emulator / distillation”, not as “independent confirmation of the physics”.

1.3.4 Result 4 — The reference scale (YOUR MAIN FIGURE)

figures/summary_scale_GK_iterative.png

	DESI	SDSS
Synthetic-trained → real	0.752	0.956
Real-trained → real (ceiling)	0.843	0.940
Domain shift	+0.090	−0.017
Cross-survey (real DESI → real SDSS)	—	0.828

Say this sentence out loud in the talk:

“On SDSS, training on nothing but simulated spectra is as good as training on real labelled data — and we used ten thousand synthetic spectra where only five hundred and ninety-four real labelled ones exist.”

1.3.5 Result 5 — Diagnosing DESI (a chain of falsified hypotheses)

This is what makes the work scientific rather than a leaderboard entry. **Four hypotheses, each tested, each refuted:**

Hypothesis	Test	Result
DESI contains giants (17 %, $\log g < 3.5$) outside the dwarf-only training grid	filter them out	NO 0.760 → 0.770
DESI contains stars outside the Teff grid	filter to 4000–6000 K, $\log g > 4$	NO 0.753
DESI spectra are not radial-velocity corrected	cross-correlation (validated by injecting known shifts)	NO measured shift < the measurement noise floor
The continuum normalization is biased by DESI’s noise	compare blue vs red envelope levels	NO no bias

Then we stopped guessing and just looked: the mean G-star spectrum, same Teff window, DESI vs SDSS. Every line agrees — **except H-beta, 20 % shallower in DESI.**

And we proved causality: masking H-beta out of the classifier lifts DESI’s G-recall from 0.67 to **0.81** — the bias vanishes exactly when the broken line is removed.

Honest negative result (include it — it strengthens you): the obvious fix, masking the lines when fitting the continuum, *does* repair H-beta (ratio 0.80 → 0.94) but **over-deepens every other line** (Ca I ratio 1.10 → 1.32) and makes DESI **worse overall** (0.752 → 0.637). We found the cause, tested the natural fix, and report why it does not hold.

1.3.6 Result 6 — Interpretability, on BOTH levels

(a) The classifier — SHAP. The Random Forest decides with the **Balmer lines**: 5 of the top 10 wavelengths are H-beta pixels (4859–4864 Å), then H-gamma. Gini importance agrees independently. Physically correct: G vs K is a temperature cut, and Balmer is the thermometer.

(b) The emulator — sensitivity analysis. We measured $\partial \text{flux} / \partial [X/H]$ for all 10 varied elements.

Element	G star	K star	Has lines in 4000–5000 Å?
Fe	1.00	1.00	yes, hundreds
C	0.46	0.61	yes (CH G-band)
Mg	0.37	0.69	yes
Ca	0.27	0.42	yes (4226.7 Å)
Ti	0.21	0.34	yes
Mn	0.20	0.22	yes
Na	0.06	0.17	no (D lines at 5890 Å)
S	0.04	0.05	no
N	0.07	0.03	no
K	0.02	0.03	no (7665 Å)

This was a falsifiable test and it passed. Elements with no lines in the window show ~zero sensitivity; the others peak *on their known lines* (Ca I exactly at 4226.7 Å). **TransformerPayne learned physics, not correlations.**

It is even subtler than that. The emulator distinguishes three regimes: - **atomic lines** → sharp, discrete peaks (Fe) - **molecular bands** → broad, diffuse elevation (C, via CH — and *stronger in the cooler K star*, because molecules survive in cool atmospheres) - **indirect continuum effects** → smooth, structureless bands (Na has *no* line here, yet rises from 0.06 to 0.17 in the K star: Na is a low-ionization **electron donor** and changes the H⁻ continuous opacity)

(c) THE PUNCHLINE — connect (a) and (b):

The emulator says Ca I and the CH G-band are controlled by **abundance**. We randomized abundances by ± 1 dex. Those lines are therefore *ambiguous* discriminators. And SHAP shows the classifier **ignores them** and uses Balmer — the only purely thermal signal, and the one element whose abundance we did *not* vary.

The classifier independently learned to avoid exactly the lines the emulator flags as abundance-contaminated.

Two independent interpretability analyses, one coherent physical picture. **This is the payoff of the project title.**

1.4 PART IV — HARD QUESTIONS AND YOUR ANSWERS

“Why only G and K? The goal was all spectral classes.” > Two independent hard limits. **(1) The emulator:** TransformerPayne is validated for dwarfs at > ~4,000–6,000 K. O (30,000 K+), B, A and M lie completely outside its training grid — it would emit > spectra, but not trustworthy ones. **(2) The data:** by the IMF, hot stars barely exist — LAMOST > contains 234 O stars in total. We also *measured* the cost of pushing outward: adding F drops the > real-data accuracy from 0.843 to 0.717 (G-recall collapses to 0.47), because F and G are adjacent > in temperature. We chose to make a rigorous claim about a validated domain rather than a weak claim > about everything.

“0.76 on DESI is not very good.” > Compared to what? DESI’s own ceiling — a model trained on real DESI data with real DESI labels — is > 0.843. And that same real-trained DESI model gets only 0.828 on real SDSS. The instruments disagree > with each other. On

SDSS our synthetic-trained model reaches 0.956 against a ceiling of 0.940. The > method is not the limitation; DESI's data is.

"Isn't training on real data the obvious thing to do?" > It is the *reference*, not a competing method — and it is exactly why we ran it. Note that on SDSS > there are only 594 labelled spectra to train on, and we used 10,000 synthetic ones. The synthetic > route gives unlimited data, needs no labels, and matches the ceiling. That is the result.

"Your Random Forest depends on a single line. Isn't that fragile?" > Yes — and we demonstrated the fragility rather than hiding it. Removing H-beta costs SDSS 14 points > (0.953 → 0.813). That single-feature dependence is precisely why DESI's 20 %-shallow H-beta is so > destructive. It is a genuine robustness finding, and the natural next step would be a model > regularized to spread its evidence across more lines.

"Why is the second emulator not an independent check?" > Correct, and we say so. Our Payne MLP is *trained on* TransformerPayne output — it is a fast > surrogate, a distillation. It cannot be better than its teacher. What it does show is that the > transformer architecture carries no advantage for this task, i.e. the emulator is not the > bottleneck.

"How do you know the emulator isn't just memorizing correlations?" > We tested it and it could have failed. Four of the ten elements we vary (Na, S, N, K) have **no > lines** in 4000–5000 Å. A correlation-learner would still show sensitivity to them. TransformerPayne > shows ~zero (0.02–0.07), while the elements that *do* have lines peak exactly on them — Ca I at > 4226.7 Å. And it reproduces subtleties nobody trained it to reproduce: molecular bands broaden in > cool stars, and Na — with no line at all here — shows a *smooth* continuum-opacity response because > it is an electron donor.

1.5 PART V — THE ONE CAVEAT TO NEVER FORGET

TransformerPayne is only valid for dwarfs at ~4,000–6,000 K. Every conclusion in this work lives inside that box. Applying the model outside it is not a bug of the method — it is a misuse of the tool. State the validity domain *before* you state the results, and the “why not all classes” question answers itself.

1.6 PART VI — SUGGESTED SLIDE ORDER

1. **The question** — can we train on simulations and work on the sky?
2. **The tools** — emulators (Payne / TransformerPayne), and their validity domain (*state it here*)
3. **The pipeline** — common grid, LSF broadening, continuum normalization, Random Forest
4. **Closing the domain shift** — 0.66 → 0.74, and *why the two effects are coupled*
5. **Scaling saturates** — the curve. More data buys nothing. (*This answers the 700k crisis.*)
6. **The reference scale** — **MAIN FIGURE**. Synthetic ≈ ceiling on SDSS. Domain shift = 0.
7. **Diagnosing DESI** — four falsified hypotheses → the H-beta measurement → causal proof → the honest negative result
8. **Interpretability I** — SHAP: the classifier uses Balmer
9. **Interpretability II** — the emulator learned real element↔line physics (*the falsifiable test*)
10. **The punchline** — the classifier avoids exactly the lines the emulator flags as abundance-contaminated
11. **Limits and next steps** — validity domain, DESI's labels, single-feature fragility

1.7 APPENDIX — NUMBERS TO HAVE ON THE TIP OF YOUR TONGUE

Wavelength range	4000–5000 Å, 1000 px
Classes	G (5300–6000 K), K (4000–5200 K), boundary 5250 K
Synthetic training set	10,000 (saturates at 2,000)
Sim → real, SDSS	0.956
Sim → real, DESI	0.752
Ceiling (real-trained), SDSS	0.940 (<i>from only 594 labelled spectra</i>)
Ceiling (real-trained), DESI	0.843
Domain shift, SDSS	−0.017 (zero)
Domain shift, DESI	+0.090
Cross-survey real DESI → real SDSS	0.828
DESI H-beta depth vs SDSS	0.80 (20 % shallower)
Cost of removing H-beta (SDSS)	0.953 → 0.813
The Payne vs TransformerPayne	0.77 vs 0.78
F/G/K on real data	0.717 (G-recall 0.47)
Emulator sensitivity, Fe	1.00
Emulator sensitivity, Na/S/N/K (no lines here)	0.02–0.07
O stars in all of LAMOST	234
